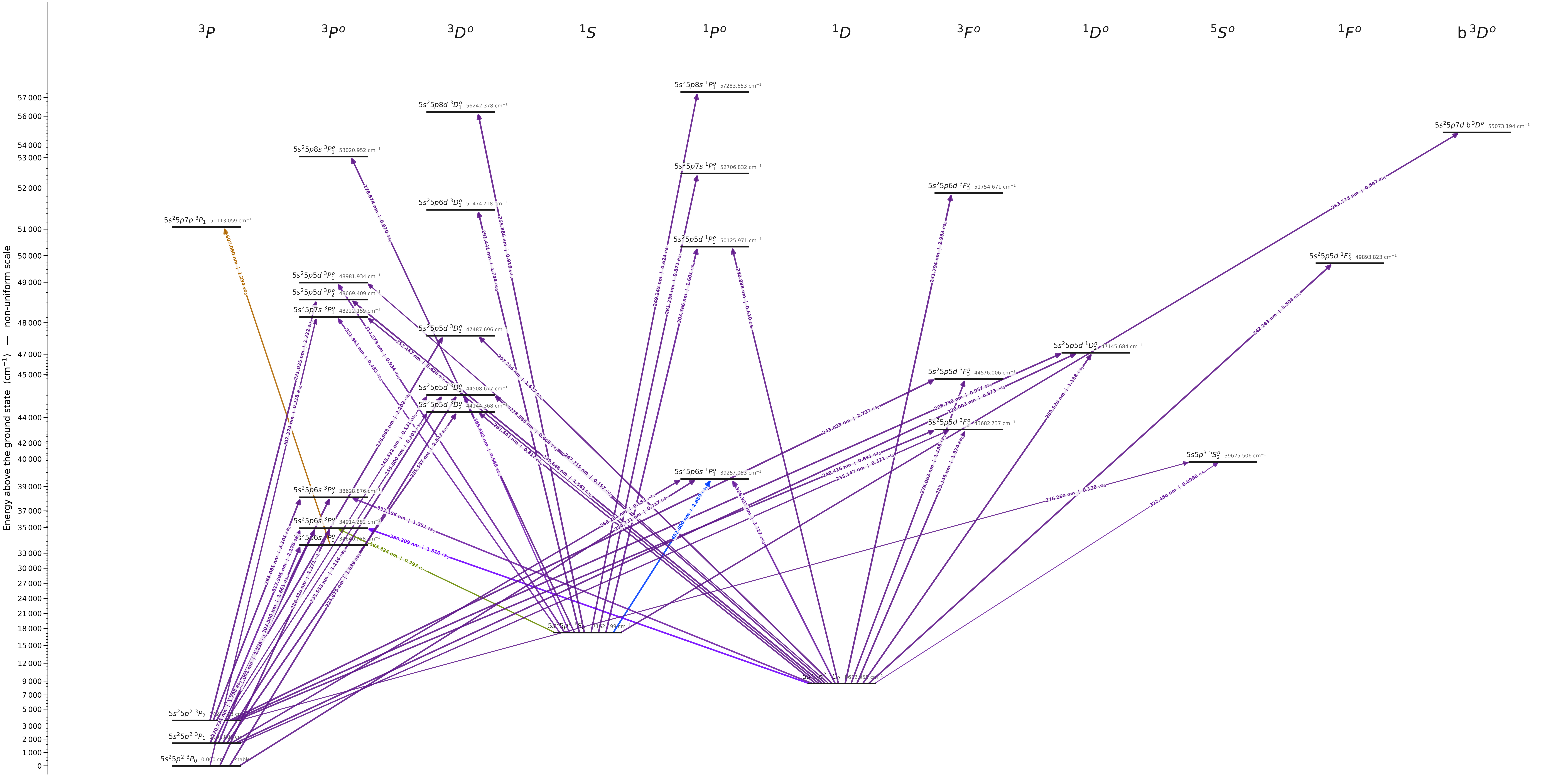
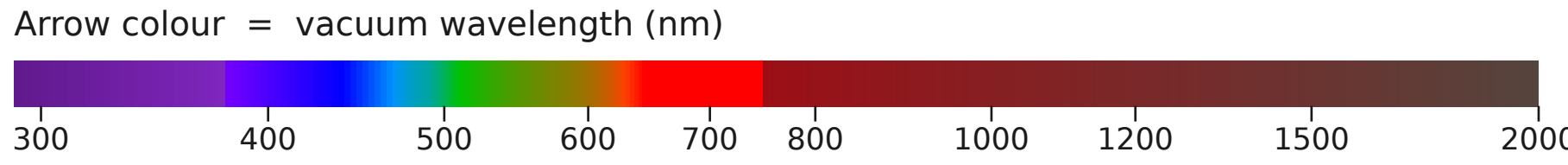


Tin

Sn (Sn I). Strongest classified transitions below 2 μm (NIST): 29 levels, 51 lines.

Sn ⁺ ionisation limit 59232.69 cm⁻¹ = 7.34392 eV (168.826 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $| \langle J || e r || J' \rangle |$ in ea₀
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, $n = 70$.
Level caption: NIST energy and measured lifetime (where one exists).



Strongest transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	⟨J er J'⟩ (ea ₀)	A (s ⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
5s²5p² ³P₂ - 5s²5p5d ³P₂°	221.0348	220.9660	1356.3131	1.222	5.600e+07	-	-	
5s²5p² ³P₀ - 5s²5p5d ³D₂°	224.6753	224.6056	1334.3366	1.639	1.600e+08	-	-	
5s²5p² ³P₂ - 5s²5p5d ³D₃°	226.9631	226.8930	1320.8863	2.202	1.200e+08	-	-	
5s²5p² ¹D₂ - 5s²5p6d ³F₃°	231.7942	231.7230	1293.3561	2.933	2.000e+08	-	-	
5s²5p² ³P₁ - 5s²5p5d ³D₂°	233.5528	233.4812	1283.6175	1.116	6.600e+07	-	-	
5s²5p² ³P₁ - 5s²5p5d ³D₂°	235.5570	235.4850	1272.6958	2.342	1.700e+08	-	-	
5s²5p² ¹D₂ - 5s²5p5d ¹F₃°	242.2430	242.1694	1237.5693	3.504	2.500e+08	-	-	
5s²5p² ³P₂ - 5s²5p5d ³F₃°	243.0232	242.9495	1233.5960	2.727	1.500e+08	-	-	
5s²5p² ¹D₂ - 5s²5p5d ³P₂°	249.6477	249.5724	1200.8623	1.543	6.200e+07	-	-	
5s²5p² ³P₁ - 5s²5p6s ³P₂°	270.7307	270.6505	1107.3455	1.798	6.600e+07	-	-	
5s²5p² ³P₂ - 5s²5p6s ³P₂°	284.0812	283.9977	1055.3055	3.101	1.700e+08	-	-	
5s²5p² ¹S₀ - 5s²5p6d ³D₁°	291.4414	291.3560	1028.6544	1.744	8.300e+07	-	-	
5s²5p² ¹S₀ - 5s²5p5d ¹P₁°	303.3661	303.2778	988.2200	1.601	6.200e+07	-	-	
5s²5p² ³P₁ - 5s²5p6s ³P₀°	303.4998	303.4115	987.7847	1.661	2.000e+08	-	-	
5s²5p² ³P₂ - 5s²5p6s ³P₁°	317.5953	317.5035	943.9448	2.178	1.000e+08	-	-	
5s²5p² ¹D₂ - 5s²5p6s ¹P₁°	326.3271	326.2330	918.6870	3.727	2.700e+08	-	-	

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ/2π: natural linewidth of the upper level, 1/(2πτ).